Applications of Inner Product Spaces

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Table of Contents

Orthogonal Matrices

Definition (Orthogonal Matrices)

Let A be a square matrix of size $n \times n$ then A is said to be orthogonal if

$$AA^T = A^T A = I_n$$

Orthogonal Matrices

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$$AA^T = A^T A = I_n$$

Example

Let $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ then

$$AA^T = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Therefore A is an orthogonal matrix.

QR factorization

QR factorization

The *QR* decomposition (also called the *QR* factorization) of a matrix A is the decomposition of the matrix A into an orthogonal matrix and a upper triangular matrix, i.e., there is a unique orthogonal matrix Q and an upper triangular matrix R with only positive numbers on its main diagonal and

$$A = QR$$

Example

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ then A has the QR factorization given below

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix} \cdot \begin{bmatrix} \sqrt{2} & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{\frac{3}{2}} \end{bmatrix}$$

Here $Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}$ is an orthogonal matrix and $R = \begin{bmatrix} \sqrt{2} & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{\frac{3}{2}} \end{bmatrix}$ is an upper triangular matrix.

Existence of QR factorization

The following result talks about the existence of QR factorization of a matrix A.

Result

Let A be a square matrix of size $n \times n$ such that its columns are linearly independent then its QR factorization exists.

How to find the QR factorization?

There are several methods for computing QR factorization of a given matrix. One of such method is Gram-Schmidt orthogonalization process.

Consider the Gram-Schmidt orthogonalization process with the vectors to be considered in the process as the columns of matrix A. That is, if

$$A = [a_1 \ a_2 \ \cdots \ a_n]$$

then

$$\begin{aligned}
u_1 &= a_1 \\
u_2 &= a_2 - \frac{\langle a_2, u_1 \rangle}{\|u_1\|^2} u_1 \\
u_3 &= a_3 - \frac{\langle a_3, u_1 \rangle}{\|u_1\|^2} u_1 - \frac{\langle a_3, u_2 \rangle}{\|u_2\|^2} u_2 \\
\vdots \quad & \qquad\quad \vdots \qquad\qquad\qquad \vdots \\
u_n &= a_n - \frac{\langle a_n, u_1 \rangle}{\|u_1\|^2} u_1 - \frac{\langle a_n, u_2 \rangle}{\|u_2\|^2} u_2 - \cdots - \frac{\langle a_n, u_{n-1} \rangle}{\|u_{n-1}\|^2} u_{n-1}
\end{aligned}$$

Cont...

Then $\{u_1, u_2, \dots, u_n\}$ is an orthogonal set of vectors. Set $v_i = \dfrac{u_i}{\|u_i\|}$ then $\{v_1, v_2, \dots, v_n\}$ is an orthonormal set of vectors. The resulting QR factorization is given by

$$A = [a_1 \ a_2 \cdots a_n] = [v_1 \ v_2 \ \cdots \ v_n] \begin{bmatrix} \langle v_1, a_1 \rangle & \langle v_1, a_2 \rangle & \cdots & \langle v_1, a_n \rangle \\ \langle v_2, a_1 \rangle & \langle v_2, a_2 \rangle & \cdots & \langle v_2, a_n \rangle \\ \vdots & \vdots & \dots & \vdots \\ \langle v_n, a_1 \rangle & \langle v_n, a_2 \rangle & \cdots & \langle v_n, a_n \rangle \end{bmatrix}$$

where $Q = [v_1 \ v_2 \ \cdots \ v_n]$ is an orthogonal matrix and

$$R = \begin{bmatrix} \langle v_1, a_1 \rangle & \langle v_1, a_2 \rangle & \cdots & \langle v_1, a_n \rangle \\ 0 & \langle v_2, a_2 \rangle & \cdots & \langle v_2, a_n \rangle \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \cdots & \langle v_n, a_n \rangle \end{bmatrix}$$

is an upper triangular matrix.

Example

Find the QR factorization of the matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.

Solution

Set $a_1 = \begin{bmatrix}1\\1\\0\end{bmatrix}$, $a_2 = \begin{bmatrix}1\\0\\1\end{bmatrix}$ and $a_3 = \begin{bmatrix}0\\1\\1\end{bmatrix}$. We'll use Gram-Schmidt orthogonalization process to find an orthogonal set of vectors.

$$u_1 = a_1 = \begin{bmatrix}1\\1\\0\end{bmatrix}$$

$$u_2 = a_2 - \frac{\langle a_2, u_1\rangle}{\|u_1\|^2}u_1 = \begin{bmatrix}\frac{1}{2}\\-\frac{1}{2}\\1\end{bmatrix}$$

$$u_3 = a_3 - \frac{\langle a_3, u_1\rangle}{\|u_1\|^2}u_1 - \frac{\langle a_3, u_2\rangle}{\|u_2\|^2}u_2 = \begin{bmatrix}-\frac{2}{3}\\\frac{2}{3}\\\frac{2}{3}\end{bmatrix}$$

Solution cont...

Thus,

$$v_1 = \frac{u_1}{\|u_1\|} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \quad v_2 = \frac{u_2}{\|u_2\|} = \begin{bmatrix} \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{3} \end{bmatrix}, \quad v_3 = \frac{u_3}{\|u_3\|} = \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$$

and hence

$$Q = [v_1 \;\; v_2 \;\; v_3] = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{\sqrt{6}}{6} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{\sqrt{6}}{6} & \frac{1}{\sqrt{3}} \\ 0 & \frac{\sqrt{6}}{3} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

and

$$R = \begin{bmatrix} \langle v_1, a_1 \rangle & \langle v_1, a_2 \rangle & \langle v_1, a_3 \rangle \\ 0 & \langle v_2, a_2 \rangle & \langle v_2, a_3 \rangle \\ 0 & 0 & \langle v_3, a_3 \rangle \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{3}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}.$$

Exercise 1:

Find the QR factorization of the matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -4 \\ 0 & 3 & 2 \end{bmatrix}$.

Relations of fundamental subspaces

- In Module-2, we discussed about four subspaces namely null space of A ($\mathcal{N}(A)$), row space of A ($\mathcal{R}(A)$), column space of A ($\mathcal{C}(A)$) and null space of A^T ($\mathcal{N}(A^T)$) of an $m \times n$ matrix A.

- One of the most important applications of the orthogonal projection of vectors onto a subspace is to study the relations or structures of the four fundamental subspaces $\mathcal{N}(A)$, $\mathcal{R}(A)$, $\mathcal{C}(A)$ and $\mathcal{N}(A^T)$ of an $m \times n$ matrix A.

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- One of the most important applications of the orthogonal projection of vectors onto a subspace is to study the relations or structures of the four fundamental subspaces $\mathcal{N}(A)$, $\mathcal{R}(A)$, $\mathcal{C}(A)$ and $\mathcal{N}(A^T)$ of an $m \times n$ matrix A.

Result

For any $m \times n$ matrix A, the null space $\mathcal{N}(A)$ and the row space $\mathcal{R}(A)$ are orthogonal in $\mathbb{R}^n$. Similarly, the null space $\mathcal{N}(A^T)$ of A^T and the column space $\mathcal{C}(A) = \mathcal{R}(A^T)$ are orthogonal in $\mathbb{R}^m$.

Result

For any $m \times n$ matrix A

① $\mathcal{N}(A) \oplus \mathcal{R}(A) = \mathbb{R}^n$.

② $\mathcal{N}(A^T) \oplus \mathcal{C}(A) = \mathbb{R}^m$

Example

Given two vectors $\begin{bmatrix}1\\2\\1\\2\end{bmatrix}$ and $\begin{bmatrix}0\\-1\\-1\\1\end{bmatrix}$, find all vectors in $\mathbb{R}^4$ that are perpendicular to them.

Solution

Suppose $\begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} \in \mathbb{R}^4$ that is perpendicular to $\begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ -1 \\ -1 \\ 1 \end{bmatrix}$, i.e.,

$$\left\langle \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \end{bmatrix} \right\rangle = 0 \quad \text{and} \quad \left\langle \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ -1 \\ 1 \end{bmatrix} \right\rangle = 0$$

$$\implies x + 2y + z + 2t = 0 \quad \text{and} \quad -y - z + t = 0$$

$$\implies y = t - z \quad \& \quad x = -4t + z$$

The solution set of above homogeneous system of linear equations is

$$= \left\{ \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} \ : \ x = -4t + z \ \& \ y = t - z \right\}$$

$$= \left\{ \begin{bmatrix} -4t + z \\ t - z \\ z \\ t \end{bmatrix} \ : \ x = -4t + z \ \& \ y = t - z \right\}$$

$$= \left\{ \begin{bmatrix} -4t \\ t \\ 0 \\ t \end{bmatrix} + \begin{bmatrix} z \\ -z \\ z \\ 0 \end{bmatrix} \ : \ t, z \in \mathbb{R} \right\}$$

$$= \left\{ t \begin{bmatrix} -4 \\ 1 \\ 0 \\ 1 \end{bmatrix} + z \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} \ : \ t, z \in \mathbb{R} \right\}$$

The above set is the collection of all the vectors that are perpendicular to given two vectors.

Exercise 2:

Find a basis for the orthogonal complement of the row space of A where

(1) $A = \begin{bmatrix} 1 & 2 & 8 \\ 2 & -1 & 1 \\ 3 & 0 & 6 \end{bmatrix}$

(2) $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Least square solutions

Recall the following result

Result

The system $Ax = b$ has atleast one solution if and only if b belongs to the column space $\mathcal{C}(A)$ of A. In this case, such a solution is unique if and only if the null space $\mathcal{N}(A)$ of A is trivial.

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Least square solution

If $b \notin \mathcal{C}(A)$ then $Ax = b$ is inconsistent. Note that for any $x \in \mathbb{R}^n$, $Ax \in \mathcal{C}(A)$. Thus, the best we can do is to find a vector $x_0 \in \mathbb{R}^n$ such that Ax_0 is closest to the given vector $b \in \mathbb{R}^m$, i.e., $\|Ax_0 - b\|$ is as small as possible. Such a solution vector x_0 gives the best approximation Ax_0 to b, and is called least square solution to $Ax = b$.

Note

However, since we have the orthogonal decomposition $\mathbb{R}^m = \mathcal{C}(A) \oplus \mathcal{N}(A^T)$, we know that for any $b \in \mathbb{R}^m$, $Proj_{\mathcal{C}(A)}(b) = b_c \in \mathcal{C}(A)$ is the closest vector to b among the vectors in $\mathcal{C}(A)$. Therefore, a least square solution $x_0 \in \mathbb{R}^n$ satisfies the following:

$$Ax_0 = b_c = Proj_{\mathcal{C}(A)}(b)$$

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Definition(Normal Equation)

Let $Ax = b$ be system of m linear equations in n unknowns. Then the equation $A^T Ax = A^T b$ is called normal equation.

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Definition(Normal Equation)

Let $Ax = b$ be system of m linear equations in n unknowns. Then the equation $A^T Ax = A^T b$ is called normal equation.

Result

Let A be $m \times n$ matrix and let $b \in \mathbb{R}^m$ be any vector. Then a vector $x_0 \in \mathbb{R}^n$ is a least square solution of $Ax = b$ if and only if x_0 is the solution of the normal equation $A^T Ax = A^T b$.

Example

Find all the least square solutions to $Ax = b$ and then determine the orthogonal projection b_c of b into the column space $\mathcal{C}(A)$ of A, where

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 2 & -3 & -1 \\ -1 & 1 & 2 \\ 3 & -5 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Solution

$$A^TA = \begin{bmatrix} 1 & 2 & -1 & 3 \\ -2 & -3 & 1 & -5 \\ 1 & -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & -2 & 1 \\ 2 & -3 & -1 \\ -1 & 1 & 2 \\ 3 & -5 & 0 \end{bmatrix} = \begin{bmatrix} 15 & -24 & -3 \\ -24 & 39 & 3 \\ -3 & 3 & 6 \end{bmatrix}$$

$$A^Tb = \begin{bmatrix} 1 & 2 & -1 & 3 \\ -2 & -3 & 1 & -5 \\ 1 & -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix}$$

A least square solution of $Ax = b$ is a solution of $A^TAx = A^Tb$ (Normal Equation), i.e.,

$$\begin{bmatrix} 15 & -24 & -3 \\ -24 & 39 & 3 \\ -3 & 3 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix}$$

The augmented matrix of above system of linear equations is

$$\left[\begin{array}{ccc|c} 15 & -24 & -3 & 0 \\ -24 & 39 & 3 & -1 \\ -3 & 3 & 6 & 3 \end{array}\right] R_1 \leftrightarrow R_3 \left[\begin{array}{ccc|c} -3 & 3 & 6 & 3 \\ -24 & 39 & 3 & -1 \\ 15 & -24 & -3 & 0 \end{array}\right] \begin{array}{l} R_1 \to \frac{R_1}{-3} \\ R_2 \to \frac{R_2}{3} \\ R_3 \to \frac{R_3}{3} \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ -8 & 13 & 1 & \frac{-1}{3} \\ 5 & -8 & -1 & 0 \end{array}\right] \begin{array}{l} R_2 \to R_2 + 8R_1 \\ R_3 \to R_3 - 5R_1 \end{array} \left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 0 & 5 & -15 & \frac{-25}{3} \\ 0 & -3 & 9 & 5 \end{array}\right] \begin{array}{l} R_2 \to \frac{R_2}{5} \\ R_3 \to \frac{R_3}{-3} \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 0 & 1 & -3 & \frac{-5}{3} \\ 0 & 1 & -3 & \frac{-5}{3} \end{array}\right] R_3 \to R_3 - R_2 \left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 0 & 1 & -3 & \frac{-5}{3} \\ 0 & 0 & 0 & 0 \end{array}\right]$$

which is in row echelon form. Here rank of coefficient matrix =rank of augmented matrix=2 $<$ 3(No. of variables). Hence this system has infinitely many solutions. The system of linear equations corresponding to above row echelon form is

$$x_1 - x_2 - 2x_3 = -1 \implies x_1 = 5x_3 - \frac{8}{3}$$

$$x_2 - 3x_3 = \frac{-5}{3} \implies x_2 = 3x_3 - \frac{5}{3}$$

Thus, the solution set is

$$= \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x_1 = 5x_3 - \frac{8}{3} \& x_2 = 3x_3 - \frac{5}{3} \right\} = \left\{ \begin{bmatrix} 5x_3 - \frac{8}{3} \\ 3x_3 - \frac{5}{3} \\ x_3 \end{bmatrix} : x_3 \in \mathbb{R} \right\}$$

$$= \left\{ \begin{bmatrix} -\frac{8}{3} \\ -\frac{5}{3} \\ 0 \end{bmatrix} + \begin{bmatrix} 5x_3 \\ 3x_3 \\ x_3 \end{bmatrix} : x_3 \in \mathbb{R} \right\} = \left\{ \begin{bmatrix} -\frac{8}{3} \\ -\frac{5}{3} \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix} : x_3 \in \mathbb{R} \right\}$$

The above set contains all the least square solution to $Ax = b$. Any least square solution of $Ax = b$ will be $x_0 = \begin{bmatrix} -\frac{8}{3} \\ -\frac{5}{3} \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 5x_3 - \frac{8}{3} \\ 3x_3 - \frac{5}{3} \\ x_3 \end{bmatrix}$ and therefore orthogonal projection of $b = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ onto $\mathcal{C}(A)$ is

$$Proj_{\mathcal{C}(A)}(b) = b_c = Ax_0 = \begin{bmatrix} 1 & -2 & 1 \\ 2 & -3 & -1 \\ -1 & 1 & 2 \\ 3 & -5 & 0 \end{bmatrix} \begin{bmatrix} 5x_3 - \frac{8}{3} \\ 3x_3 - \frac{5}{3} \\ x_3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Example

Find the equation of the straight line that best fits the data of the four points data $(1,0),(2,3),(3,4)$ and $(4,4)$.

Solution

The general equation for a line is

$$y = ax + b$$

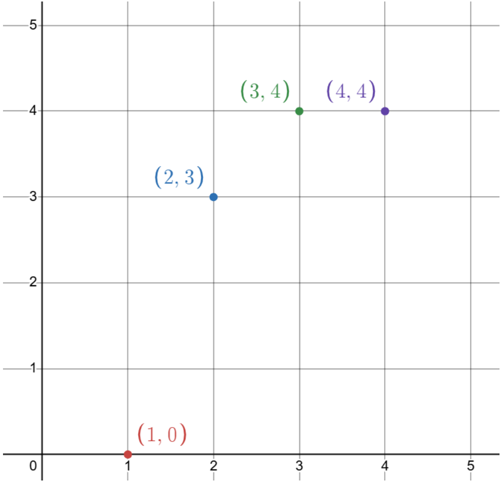
If four data points were to lie on this line, then the following equations would be satisfied:

$$0 = m + c$$
$$3 = 2m + c$$
$$4 = 3m + c$$
$$4 = 4m + c$$

In order to find the best-fit line, we try to solve the above equations in the unknowns m and c. As the four points do not actually lie on a line, there is no actual solution, so instead we compute a least-squares solution.

Putting our linear equations into matrix form, we are trying to solve $Ax = b$ for

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \qquad x = \begin{bmatrix} m \\ c \end{bmatrix} \qquad b = \begin{bmatrix} 0 \\ 3 \\ 4 \\ 4 \end{bmatrix}.$$



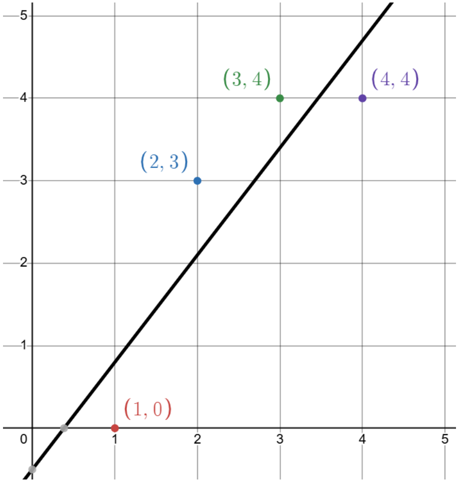
We have

$$A^TA = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix} \text{ and } A^Tb = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 34 \\ 11 \end{bmatrix}$$

The augmented matrix of $A^T Ax = A^T b$ is

$$\left[\begin{array}{cc|c} 30 & 10 & 34 \\ 10 & 4 & 11 \end{array}\right] \xrightarrow{RREF} \left[\begin{array}{cc|c} 1 & 0 & 1.3 \\ 0 & 1 & -0.5 \end{array}\right]$$

This gives $m = 1.3$ and $c = -0.5$. Thus equation of best fit line is $y = 1.3x - 0.5$.



Exercise 3:

Find all least square solutions x in $\mathbb{R}^3$ of $Ax = b$. Also, find the projection of b onto $\mathcal{C}(A)$, where

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & -1 \\ -1 & 2 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} 3 \\ -3 \\ 0 \\ -3 \end{bmatrix}$$

Exercise 4:

Find

(a) a least-squares sollution of $Ax = b$

(b) the orthogonal projection of $b = \begin{bmatrix} 2 \\ 5 \\ 6 \\ 6 \end{bmatrix}$,

where $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \\ -1 & 1 & -1 \end{bmatrix}$.

Exercise 5:

Find a least square solution to the following system of linear equations

$$\begin{bmatrix} 3 & -6 \\ 4 & -8 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 7 \\ 2 \end{bmatrix}$$

Exercise 6:

Find a least square solution to the following system of linear equations

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & 2 & -3 \end{bmatrix}, \qquad b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$$

and hence find orthogonal projection of b onto $\mathcal{C}(A)$.

Exercise 7:

Find the least square solutions of $Ax = b$, where

$$A = \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Exercise 8:

Find the least square solutions of $Ax = b$, where

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$$

Exercise 9:

Find the equation of the straight line that best fits the data of the three points data $(0, 6), (1, 0), (2, 0)$.

Exercise 10:

Find the parabola that best approximates the data points $(-1, 1/2), (1, -1), (2, -1/2), (3, 2)$.

Exercise 11:

Find the cubic polynomial that best fits the data of the five points $(-1, -14), (0, -5), (1, -4), (2, 1)$, and $(3, 22)$.

Singular value decomposition

Recall

- All eigenvalues of a real symmetric matrix are real.
- Eigenvectors of a real symmetric matrix associated with distinct eigenvalues are orthogonal.

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- Eigenvectors of a real symmetric matrix associated with distinct eigenvalues are orthogonal.

Remark

Let A be an $m \times n$ matrix. Consider the matrix $A^T A$. Then it is a symmetric matrix of size $n \times n$, so its eigenvalues are real.

Result

Let A be an $m \times n$ matrix. If λ is an eigenvalue of $A^T A$, then $\lambda \geq 0$.

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Let A be an $m \times n$ matrix. If λ is an eigenvalue of $A^T A$, then $\lambda \geq 0$.

Proof

Let v be an eigenvector of $A^T A$ corresponding to the eigenvalue λ, i.e., $A^T A v = \lambda v$. We compute that

$$\|Av\|^2 = \langle Av, Av \rangle = (Av)^T Av = v^T A^T A v = v^T \lambda v = \lambda v^T v = \lambda \langle v, v \rangle = \lambda \|v\|^2$$

Since $\|Av\|^2 \geq 0$, it follows from the above equation that $\lambda \|v\|^2 \geq 0$. Since $\|v\|^2 > 0$ (as our convention is that eigenvectors are non-zero), we deduce that $\lambda \geq 0$.

Definition(Singular values)

Let A be an $m \times n$ matrix. Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ denote the eigenvalues of $A^T A$, with repetitions. Order these so that $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$. Let $\sigma_i = \sqrt{\lambda_i}$, so that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$. The numbers $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_n \geq 0$ are called the singular values of A.

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Result

The number of non-zero singular values of A equals the rank of A.

Example

Find the singular values of the matrix $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix}$.

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Find the singular values of the matrix $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix}$.

Solution

Here $rank(A)$ is 2 and hence A has two singular values. Now

$$A^T A = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$$

The characteristics polynomial of $A^T A$ is

$$p_{A^T A}(\lambda) = det(A^T A - \lambda I_2) = \begin{vmatrix} 17 - \lambda & 8 \\ 8 & 17 - \lambda \end{vmatrix} = \lambda^2 - 34\lambda + 255 = (\lambda - 25)(\lambda - 9)$$

Thus, the eigenvalues of $A^T A$ are $\lambda_1 = 25$ and $\lambda_2 = 9$. So the singular are $\sigma_1 = \sqrt{25} = 5$ and $\sigma_2 = \sqrt{9} = 3$.

Definition(Singular value decomposition)

Let A be an $m \times n$ matrix. A singular value decomposition of A is a factorization

$$A = U\Sigma V^T$$

where:

- U is an $m \times m$ orthogonal matrix.
- V is an $n \times n$ orthogonal matrix.
- Σ is an $m \times n$ matrix whose ith diagonal entry equals the ith singular value i for $i = 1, 2, \ldots, r$, where r is the rank of A. All other entries of are zero.

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Remark

The columns of matrix V are called right singular vectors and the columns of matrix U are called left singular vectors.

Result

Let A be an $m \times n$ matrix and $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$ be singular values of A. Let v_i be an orthonormal eigenvector of $A^T A$ corresponding to eigenvalue ${\sigma_i}^2$. Then

① $\|Av_i\| = \sigma_i$.

② If $i \neq j$ then Av_i and Av_j are orthogonal.

Result

Let A be an $m \times n$ matrix and $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$ be singular values of A. Let v_i be an orthonormal eigenvector of A^TA corresponding to eigenvalue ${\sigma_i}^2$. Then

1. $\|Av_i\| = \sigma_i$.
2. If $i \neq j$ then Av_i and Av_j are orthogonal.

Result

Let A be an $m \times n$ matrix. Then A has a (not unique) singular value decomposition $A = U\Sigma V^T$,where U and V are as follows:

- The columns of V are orthonormal eigenvectors $v_1, \ldots v_n$ of A^TA, where $A^TAv_i = {\sigma_i}^2 v_i$.
- If $i \leq r$, so that $\sigma_i \neq 0$, then the ith column u_i of U is ${\sigma_i}^{-1}Av_i$. By above result, these columns are orthonormal, and the remaining columns of U are obtained by arbitrarily extending to an orthonormal basis for $\mathbb{R}^m$.

Remark

We know that the v_i's are eigenvectors of A^TA corresponding to eigenvalues ${\sigma_i}^2$,i.e.,

$$\begin{aligned} A^TAv_i &= {\sigma_i}^2 v_i \\ \implies (AA^T)Av_i &= {\sigma_i}^2 Av_i \\ \implies AA^T(\sigma_i u_i) &= {\sigma_i}^2 \sigma_i u_i \\ AA^T u_i &= {\sigma_i}^2 u_i \end{aligned}$$

This shows that u_i is an eigenvector of AA^T corresponding to eigenvalue ${\sigma_i}^2$. Thus, columns u_i of U are orthonormal eigenvectors of AA^T corresponding to eigenvalue ${\sigma_i}^2$ and columns of V are orthonormal eigenvectors of V corresponding to eigenvalue ${\sigma_i}^2$.

Steps for Calculation of singular value decomposition

- **Step-1:** Compute $A^T A$ of a real $m \times n$ matrix A of rank r.

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Steps for Calculation of singular value decomposition

- **Step-1:** Compute A^TA of a real $m \times n$ matrix A of rank r.

- **Step-2:** Compute the eigenvalues of A^TA and hence compute singular values of A^TA.

- **Step-3:** Sort the singular values, possibly renaming them, so that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$.

Steps for Calculation of singular value decomposition

- **Step-1:** Compute $A^T A$ of a real $m \times n$ matrix A of rank r.
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- **Step-3:** Sort the singular values, possibly renaming them, so that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$.
- **Step-4:** Construct the matrix of size $m \times n$ such that $\Sigma_{ii} = 1$ for $i = 1, \ldots, r$ and $\Sigma_{ij} = 0$ when $i \neq j$.

Steps for Calculation of singular value decomposition

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- **Step-5:** Compute the eigenvectors of $A^T A$.

❁ **Step-6:** Compute the (right singular) vectors $v_1, \ldots, v_r$ by normalizing each eigenvector x_i by dividing it by $\|x_i\|$. That is, let

$$v_i = \frac{x_i}{\|x_i\|}, \quad i = 1, \ldots, r$$

If $n > r$, the additional $n - r$ vectors $v_{r+1}, \ldots, v_n$ need to be chosen as an orthonormal basis in $Null(A)$.

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- **Step-8:** Compute the (left singular) vectors $u_1, \dots, u_r$ as

$$Av_i = \sigma_i u_i \implies u_i = \frac{Av_i}{\sigma_i} = \sigma_i{}^{-1} Av_i, \quad i = 1, \dots, r$$

If $m > r$, the additional $m - r$ vectors $u_{r+1}, \dots, u_m$ need to be chosen as an orthonormal basis in $Null(A^T)$.

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If $m > r$, the additional $m - r$ vectors $u_{r+1}, \ldots, u_m$ need to be chosen as an orthonormal basis in $Null(A^T)$.

❁ **Step-9:** Construct the orthogonal matrix $U = [u_1 \ u_2 \cdots u_m]$.

Example

Find the singular value decomposition of $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$.

Example

Find the singular value decomposition of $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$.

Solution

Here $rank(A) = 2$, so A has two singular values. Now

$$A^T A = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$$

The characteristic polynomial of matrix $A^T A$ is

$$p_{A^T A}(\lambda) = det(A^T A - \lambda I_2) = \begin{vmatrix} 25 - \lambda & 20 \\ 20 & 25 - \lambda \end{vmatrix} = (\lambda - 45)(\lambda - 5) \implies \begin{array}{c} \lambda_1 = 45 \\ \lambda_2 = 5 \end{array}$$

The singular values of A are $\sigma_1 = \sqrt{45}$ and $\sigma_2 = \sqrt{5}$. This implies

$$\Sigma = \begin{bmatrix} \sqrt{45} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$$

Eigenvector of A^TA corresponding to eigenvalue $\lambda_1 = 45$: Consider

$$(A^TA - \lambda_1 I_2)X = 0 \implies \begin{bmatrix} -20 & 20 \\ 20 & -20 \end{bmatrix} X = 0$$

which gives $-20x_1 + 20x_2 = 0$, i.e., $x_1 = x_2$. Thus, eigenvector corresponding to eigenvalue λ_1 is $s_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Therefore

$$v_1 = \frac{s_1}{\|s_1\|} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

Eigenvector of A^TA corresponding to eigenvalue $\lambda_1 = 45$: Consider

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Eigenvector of A^TA corresponding to eigenvalue $\lambda_2 = 5$: Consider

$$(A^TA - \lambda_2 I_2)X = 0 \implies \begin{bmatrix} 20 & 20 \\ 20 & 20 \end{bmatrix} X = 0$$

which gives $20x_1 + 20x_2 = 0$, i.e., $x_1 = -x_2$. Thus, eigenvector corresponding to eigenvalue λ_2 is $s_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Therefore

$$v_2 = \frac{s_2}{\|s_2\|} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

Therefore

$$v = [v_1 \quad v_2] = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Therefore

$$v = [v_1 \quad v_2] = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Now compute u_1 and u_2 as following

$$u_1 = {\sigma_1}^{-1} A v_1 = \frac{1}{\sqrt{45}} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix}$$

$$u_2 = {\sigma_2}^{-1} A v_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -\frac{3}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} \end{bmatrix}$$

Therefore

$$U = [u_1 \quad u_2] = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}$$

Therefore

$$v = [v_1 \quad v_2] = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Now compute u_1 and u_2 as following

$$u_1 = {\sigma_1}^{-1} A v_1 = \frac{1}{\sqrt{45}} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix}$$

$$u_2 = {\sigma_2}^{-1} A v_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -\frac{3}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} \end{bmatrix}$$

Therefore

$$U = [u_1 \quad u_2] = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}$$

Thus

$$A = U\Sigma V^T = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix} \begin{bmatrix} \sqrt{45} & 0 \\ 0 & \sqrt{5} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Note

Notice that when A is a symmetric matrix, $A^T A = AA^T$, so $U = V$. Less work is required.

Example

Find the singular value decomposition of $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$.

Example

Find the singular value decomposition of $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$.

Solution

Note that A is a symmetric matrix so $U = V$. Now

$$A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

Here $rank(A) = 2$, so A has two singular values. The characteristics polynomial of $A^T A$ is

$$p_{A^T A}(\lambda) = det(A^T A - \lambda I_2) = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = \lambda^2 - 3\lambda + 1 \implies \begin{matrix} \lambda_1 = \frac{3+\sqrt{5}}{2} \\ \lambda_2 = \frac{3-\sqrt{5}}{2} \end{matrix}$$

Solution cont...

The singular values of A are

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{\frac{3+\sqrt{5}}{2}} = \frac{1+\sqrt{5}}{2}$$
$$\sigma_2 = \sqrt{\lambda_1} = \sqrt{\frac{3-\sqrt{5}}{2}} = \frac{1-\sqrt{5}}{2}$$

This implies

$$\Sigma = \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{bmatrix}$$

Eigenvector of A^TA corresponding to eigenvalue $\lambda_1 = \frac{3+\sqrt{5}}{2}$: Consider

$$(A^TA - \lambda_1 I_2)X = 0 \implies \begin{bmatrix} \frac{1-\sqrt{5}}{2} & 1 \\ 1 & -\frac{1+\sqrt{5}}{2} \end{bmatrix} X = 0$$

The augmented matrix of this system of linear equation is

$$\left[\begin{array}{cc|c} \frac{1-\sqrt{5}}{2} & 1 & 0 \\ 1 & -\frac{1+\sqrt{5}}{2} & 0 \end{array}\right] R_2 \to \frac{1-\sqrt{5}}{2} R_2 \left[\begin{array}{cc|c} \frac{1-\sqrt{5}}{2} & 1 & 0 \\ \frac{1-\sqrt{5}}{2} & 1 & 0 \end{array}\right] R_2 \to R_2 - R_1 \left[\begin{array}{cc|c} \frac{1-\sqrt{5}}{2} & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

The system corresponding to above REF is

$$\left(\frac{1-\sqrt{5}}{2}\right) x_1 + x_2 = 0 \implies x_2 = -\left(\frac{1-\sqrt{5}}{2}\right) x_1$$

Thus eigenvector corresponding to eigenvalue λ_1 is $s_1 = \begin{bmatrix} 1 \\ -\frac{1-\sqrt{5}}{2} \end{bmatrix}$. Thus

$$v_1 = \frac{s_1}{\|s_1\|} = \frac{2}{\sqrt{10 - 2\sqrt{5}}} \begin{bmatrix} 1 \\ -\frac{1-\sqrt{5}}{2} \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{10-2\sqrt{5}}} \\ \frac{1}{\sqrt{2\sqrt{5}}} \end{bmatrix}$$

Eigenvector of $A^T A$ corresponding to eigenvalue $\lambda_2 = \frac{3-\sqrt{5}}{2}$: Consider

$$(A^T A - \lambda_2 I_2)X = 0 \implies \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 1 \\ 1 & \frac{-1+\sqrt{5}}{2} \end{bmatrix} X = 0$$

The augmented matrix of this system of linear equation is

$$\left[\begin{array}{cc|c} \frac{1+\sqrt{5}}{2} & 1 & 0 \\ 1 & \frac{-1+\sqrt{5}}{2} & 0 \end{array}\right] R_2 \to \tfrac{1+\sqrt{5}}{2} R_2 \left[\begin{array}{cc|c} \frac{1+\sqrt{5}}{2} & 1 & 0 \\ \frac{1+\sqrt{5}}{2} & 1 & 0 \end{array}\right] R_2 \to R_2 - R_1 \left[\begin{array}{cc|c} \frac{1+\sqrt{5}}{2} & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

The system corresponding to above REF is

$$\left(\frac{1+\sqrt{5}}{2}\right) x_1 + x_2 = 0 \implies x_2 = -\left(\frac{1+\sqrt{5}}{2}\right) x_1$$

Thus eigenvector corresponding to eigenvalue λ_2 is $s_2 = \begin{bmatrix} 1 \\ -\frac{1+\sqrt{5}}{2} \end{bmatrix}$. Thus

$$v_2 = \frac{s_2}{\|s_2\|} = \frac{2}{\sqrt{10+2\sqrt{5}}} \begin{bmatrix} 1 \\ -\frac{1+\sqrt{5}}{2} \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{10+2\sqrt{5}}} \\ -\frac{1}{\sqrt{2\sqrt{5}}} \end{bmatrix}$$

Therefore

$$V = [v_1 \ \ v_2] = \begin{bmatrix} \frac{2}{\sqrt{10-2\sqrt{5}}} & \frac{2}{\sqrt{10+2\sqrt{5}}} \\ \frac{1}{\sqrt{2\sqrt{5}}} & -\frac{1}{\sqrt{2\sqrt{5}}} \end{bmatrix} = U$$

Thus

$$A = U\Sigma V^T = \begin{bmatrix} \frac{2}{\sqrt{10-2\sqrt{5}}} & \frac{2}{\sqrt{10+2\sqrt{5}}} \\ \frac{1}{\sqrt{2\sqrt{5}}} & -\frac{1}{\sqrt{2\sqrt{5}}} \end{bmatrix} \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{10-2\sqrt{5}}} & \frac{2}{\sqrt{10+2\sqrt{5}}} \\ \frac{1}{\sqrt{2\sqrt{5}}} & -\frac{1}{\sqrt{2\sqrt{5}}} \end{bmatrix}^T$$

Exercise 12:

Find the singular value decomposition of $A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$.

Exercise 13:

Find the singular value decomposition of $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$.

Exercise 14:

Find the singular value decomposition of $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$.

Exercise 15:

Find the singular value decomposition $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ \sqrt{2} & -\sqrt{2} \end{bmatrix}$.